

Gwendolyn Noria

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Flight-critical Avionics | Real-time Simulations | High-performance Graphics

Education

The Ohio State University

B.S. Computer Science and Engineering

Columbus OH

Expected May 2028

Technical Skills

Proficient: C/C++, Zig, Java, Vulkan, OpenGL, Linux, Familiar: Python, RenderDoc, QEMU, LibVirt

Experience

Avionics Lead Software Engineer

December 2025 - Present

Buckeye Space Launch Initiative (BSLI)

Columbus, OH

- Led Avionics software development for a multistage rocket targeting 45,000ft apogee within a multidisciplinary engineering team.
- Verified and Validated Air brake Control Systems through MIL, SIL, PIL, and HIL industry testing.
- Architected our firmware to leverage parallelism enabling 500hz data acquisition and 100hz control loops.
- Engineered data pipeline to enable full post-flight state reconstruction ensuring reproduction of failure modes.
- Designed fault-tolerant logging system to capture and diagnose in-flight failures.
- Managed the development of rendering applications for viewing, processing, and ingesting live telemetry and video for remote backups and intuitive statistics on our live broadcast.
- Produced documentation throughout project timelines to reduce friction and increase bus factor.
- Technologies used: C/C++, Zig, FreeRTOS, RP2350, and Arduino.

Ceresvoxel (<https://codeberg.org/gwn/ceresvoxel>)

Feb 2025 - Present

Personal Project: Video Game/Game Engine

- A video game computing real-time simulation of vehicle dynamics and factory manufacturing pipelines.
- Implemented realtime physics simulation based on Extended Position Based Dynamics (XPBD).
- Developed custom AABB collision detection and manifold creation for voxel based rigid bodies.
- Implemented Orbital Mechanics using Patched Conics and Keplerian orbits.
- Achieved no memory allocations during rendering improving frame times by 4x.
- Optimized rendering by using push constants for chunk indexing on the GPU and using multidraw calls.
- Technologies used: Zig, C/C++, Vulkan, and RenderDoc.

Computer Engineering Capstone Mentor

Jan 2026 - Present

The Ohio State University ECE Department

Columbus, OH

- Mentored senior undergraduates in firmware and embedded systems design.
- Facilitated meeting deliverable deadlines for several project teams.
- Prepared incoming students for the challenges of project management.

Peak Surfer Intern

March 2026 - Present

National Security Agency & The Ohio State University ECE Department

Columbus, OH

- Provided specifications for firmware deliverables and provided gold-standard references for ciphers.
- Mentored 4 students to implement cryptographic ciphers on FPGAs and Software Defined Radios.
- Technologies used: 32 bit KAJU cipher and 128bit SIMON cipher

The Embedded Security Club Vice President (former)

Mitre Embedded Capture the Flag (Mitre eCTF)

- Placed 9th among 80 teams in 2023.
- Defended against Carnegie Mellon's PPP without private key compromise during competition.
- Implemented key exchanges and measures to mitigate hardware access of critical systems